Distributed Key-Value Store Partitioning Lab
Setup and Execution Instructions

(on Ubuntu / Debian Linux)

1. Extract the project

Extract Distributed_KV_Partitioning_Lab_Code.zip. Open a terminal and navigate to the project directory. For example:

```
cd ~/Downloads/Distributed_KV_Partitioning_Lab_Code/Distributed_KV_Partitioning_Lab
```

Check that the directory contains:

```
app.py
load_test.py
requirements.txt
README.md
solution/
```

2. Install virtual-environment support

On Ubuntu or Debian, install Python virtual-environment support once:

```
sudo apt update
sudo apt install python3-venv
```

3. Create the virtual environment (once only)

Run this command from the project directory:

```
python3 -m venv venv
```

This creates a directory named venv inside the project directory. Do not recreate it for each node or terminal.

4. Activate the environment and install dependencies

```
source venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

The requirements.txt file should contain two separate lines:

```
Flask>=3.0,<4.0
requests>=2.31,<3.0
```

Create the environment and install its dependencies only once. Each new terminal must activate the existing environment separately, but does not need to reinstall the packages.

5. Start the three KV-store nodes

The lab runs three independent Flask processes on the same machine, using ports 5001, 5002, and 5003. Open three separate terminal windows. In each terminal, navigate to the project directory, activate the existing environment, and start the corresponding node.

Terminal 1 — Node 1

```
cd ~/Downloads/Distributed_KV_Partitioning_Lab_Code/Distributed_KV_Partitioning_Lab
source venv/bin/activate
PORT=5001 NODE_ID=node-1 python app.py
```

Terminal 2 — Node 2

```
cd ~/Downloads/Distributed_KV_Partitioning_Lab_Code/Distributed_KV_Partitioning_Lab
source venv/bin/activate
PORT=5002 NODE_ID=node-2 python app.py
```

Terminal 3 — Node 3

```
cd ~/Downloads/Distributed_KV_Partitioning_Lab_Code/Distributed_KV_Partitioning_Lab
source venv/bin/activate
PORT=5003 NODE_ID=node-3 python app.py
```

Keep all three terminals open while running the experiment. Each terminal should display that its Flask server is running on the configured port.

6. Verify that all three nodes are running

Open a fourth terminal, navigate to the project directory, and activate the environment:

```
cd ~/Downloads/Distributed_KV_Partitioning_Lab_Code/Distributed_KV_Partitioning_Lab
source venv/bin/activate
```

Check each node's health endpoint:

```
curl http://127.0.0.1:5001/health
curl http://127.0.0.1:5002/health
curl http://127.0.0.1:5003/health
```

Each request should return a JSON response with status ok and the corresponding node name (node-1, node-2, or node-3).

7. Run the baseline experiment

With all three nodes running, execute the load test from Terminal 4:

```
python load_test.py
```

The default workload uses uniformly selected keys. The script reports successful reads, errors, mean latency, median latency, 95th-percentile latency, and per-node key/read/write/forwarding metrics.

8. Run the skewed workload

```
python load_test.py --distribution skewed
```

The skewed workload sends 80% of requests to a small subset of hot keys. Compare its results with the uniform workload.

9. What to expect from the starter implementation

The starter version of app.py contains:

```
def owner_index(key):
    return 0
```

This is intentional. Initially, every key is assigned to Node 1. Nodes 2 and 3 are running, but they will not own keys in the baseline experiment. You will then modify owner_index() to implement deterministic hash-based partitioning, then rerun the workloads and compare the distribution of keys and requests.

10. Troubleshooting

Problem	Likely cause	What to do
externally-managed-environment	Trying to install packages into system Python.	Create and activate venv; install packages inside it.
venv/bin/activate: No such file	The venv directory is absent from the current project directory.	From the project directory, run python3 -m venv venv once.
No module named requests	The terminal is using a Python interpreter without the lab dependencies.	Run source venv/bin/activate, then use python app.py.
Connection refused on port 5002 or 5003	The corresponding node is not running.	Start Node 2 or Node 3 in its own terminal.
Address already in use	Another process is using the configured port.	Stop the process using that port or configure a different unused port consistently.

11. Important reminders

- Create the virtual environment only once, from the project directory.
- Install dependencies only once in that environment.
- Activate the existing environment separately in every terminal.
- Use python app.py after activation so the environment's interpreter is used.
- Run each node in a separate terminal and keep those terminals open.
- Run load_test.py only after all three nodes are running.

Lab architecture: Three independent Flask processes, each maintaining its own in-memory dictionary and listening on a separate port of the same machine. This setup supports experiments with partitioning and request routing without requiring multiple physical machines, replication, or consensus.